CrunchBite — Domain Context

About the Organisation

CrunchBite is a fast-casual QSR (Quick Service Restaurant) chain operating **680 corporate stores** and **320 franchise locations** across North America, with international expansion underway in the UK and Australia. The brand serves approximately **2.4 million transactions per day** across dine-in, drive-through, mobile order-ahead, and delivery channels.

CrunchBite's digital division — **CrunchBite Digital** — owns the customer-facing mobile app, web ordering platform, loyalty programme, and the in-store digital experience (kiosks, kitchen display systems, digital menu boards).

Business Domain Concepts

User Personas

Persona	Description	Key Needs
Guest (unauthenticated)	Walk-in customer ordering at kiosk or counter	Fast ordering, accurate order, minimal friction
Member	Loyalty programme participant	Points earn/redeem, personalised offers, order history
Franchise Owner	Owner of 1–15 franchise stores	Store performance dashboards, labour/food cost alerts
Store Manager	Manages one corporate store	Shift scheduling, void/refund controls, end-of-day reports
Kitchen Staff	Prepares orders	Clear order queue, modifier visibility, bump/recall
Drive-Through Attendant	Takes and hands off drive-through orders	Queue time, order status, upsell prompts
Marketing Manager	Corporate marketing team	Campaign management, segment targeting, A/B test offers
Franchise Business Consultant (FBC)	CrunchBite field rep supporting franchisees	Multi-store roll-up dashboards, compliance scoring

Ordering Channels

- **Mobile App (Order-Ahead)** — iOS and Android; pickup or delivery; supports scheduled orders
- **Web Ordering** — same menu and cart as mobile; primarily desktop/tablet
- **In-Store Kiosk** — Windows-based JAWS-accessible touch kiosk; 30-second idle reset
- **Drive-Through (POS at window)** — Oracle MICROS POS; licence plate recognition (LPR) integration (Phase 2)
- **Delivery Aggregators** — Uber Eats, DoorDash, Grubhub via middleware (Otter); orders injected into kitchen as if native

- **Catering / Group Orders** — minimum 10 items, 4-hour advance notice, assigned to catering coordinator role

Menu & Catalogue

- Menu items organised by **daypart**: Breakfast (open–10:30 AM), All-Day, Late Night (10 PM–close)
- **Limited Time Offers (LTOs)** scheduled by corporate; activate/deactivate by date and store
- **86'ing** (item unavailability): store manager marks item unavailable; propagates to all channels within 60 seconds
- Modifier groups: required (size, protein) and optional (sauces, toppings); max 12 modifiers per item
- Nutritional and allergen data attached to every menu item; mandatory display on kiosk and web
- **Price book** managed centrally; franchise stores can override price within ±10% band

Loyalty Programme — CrunchRewards

- **Earn**: 10 points per $1 spent (excluding tax, fees, alcohol)
- **Redeem**: points exchanged for reward tiers ($2 off / free item / limited merchandise)
- **Tiers**: Bronze (0–499 pts), Silver (500–1999 pts), Gold (2000+ pts); tier re-evaluated monthly
- **Offers**: personalised via Braze; segments by tier, visit frequency, favourite items, location
- **Referral**: member earns 200 bonus points when referred friend places first order
- Points expire after **12 months** of inactivity
- **Fraud rules**: maximum 3 points transactions per day per account; LPR-linked duplicate detection (Phase 2)

Kitchen Operations

- Orders flow to **Kitchen Display System (KDS)** — Epson KDS hardware running CrunchBite KDS app
- Order shown as **tickets** grouped by station: grill, fry, sandwich, beverage
- **Make time target**: 3 minutes for standard orders, 5 minutes for catering items
- **Expediter screen** consolidates all station outputs for order assembly
- **Bump**: kitchen staff taps item when ready; ticket clears when all items bumped
- **Recall**: last 20 bumped tickets accessible for re-display
- **Speed of Service (SOS)**: order-placed → order-ready timer shown to drive-through attendant

Drive-Through

- Dual-lane drive-through at 240 of the 680 corporate stores
- Average drive-through transaction time target: **3 minutes 30 seconds** (order-to-window)
- **AI suggestive selling**: upsell prompt shown on Order Confirmation Unit (OCU) based on order content and daypart
- **Order Confirmation Unit (OCU)**: customer-facing display at order point; shows cart in real time

Delivery

- Delivery orders fulfilled by Uber Eats and DoorDash drivers; aggregated via **Otter** middleware
- Otter injects delivery orders into Oracle MICROS POS as type `DELIVERY`
- Estimated delivery time displayed in app; sourced from aggregator API
- **Delivery radius**: configured per store (default 3 miles); can be reduced during peak/rush

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Compliance & Regulatory Context

PCI DSS v4.0

- Cardholder data handled exclusively by **Adyen** (payment gateway); CrunchBite systems are SAQ-A (no card data on own servers)

- Tokenisation: Adyen returns a reusable network token per saved card; stored in Loyalty Service
- 3DS2 required for all card-not-present transactions above $25

CCPA / US State Privacy Laws

- Members may request data export, correction, or deletion
- Marketing opt-out must be honoured within 15 business days
- "Do Not Sell or Share My Personal Information" link required on all web properties

Allergen & Nutritional Labelling

- US: FDA menu labelling rule — calorie counts mandatory on menus with >20 locations
- UK: Natasha's Law — full ingredient and allergen list mandatory on all packaged items
- Per-item allergen data sourced from Product Information Management (PIM) system; displayed on kiosk, app, and web

Accessibility

- WCAG 2.1 AA on web and mobile app
- ADA kiosk compliance: accessibility mode activates via physical button; JAWS-compatible
- UK: Equality Act 2010 — equivalent standard for UK digital properties

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Business Rules

1. A member may apply **one offer per order**; offers are single-use and expire on the campaign end date
2. Points earned on an order are **held for 24 hours** before posting (fraud hold); refunded orders forfeit points
3. Store managers may void orders up to **$75 without approval**; above $75 requires district manager PIN
4. **86-ing** an item propagates to all channels within **60 seconds** via the Menu Service event
5. Kiosk sessions idle for **30 seconds** show a soft prompt; idle for **60 seconds** reset to the attract screen
6. Delivery orders received **after store close time minus 15 minutes** are auto-rejected with guest notification
7. Catering orders require **4-hour lead time** and a minimum of **10 items**; assigned automatically to the catering coordinator on duty
8. A franchise store's price override must stay within **±10%** of the corporate price book value

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Integration Landscape

System	Purpose	Protocol
Oracle MICROS POS	In-store point of sale, order management	MICROS REST API / TCP socket
Adyen	Payment processing (all channels)	Adyen REST + Terminal API
Braze	Customer engagement, push/email/SMS campaigns	REST API + Braze SDK

Otter	Delivery aggregator middleware (Uber Eats, DoorDash, Grubhub)	REST API (webhook inbound)
Epson KDS	Kitchen display system hardware	WebSocket (CrunchBite KDS app)
Revel Systems	Kiosk software layer (wraps MICROS)	REST API
Infor CloudSuite Food & Beverage	Inventory, procurement, recipe management (ERP)	REST API + batch file export
Medallia	Guest satisfaction survey and NPS	REST API (post-transaction trigger)
Tableau / Snowflake	Analytics, franchise dashboards	JDBC / Snowflake Connector
Google Maps Platform	Store locator, delivery radius, ETA	REST API